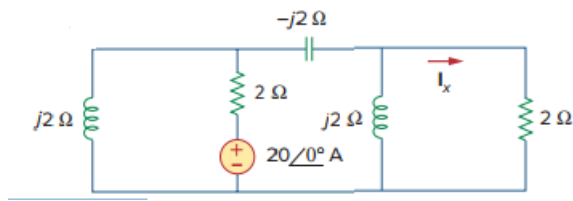


Problem

Find current I_o in the network of Fig. 9.52.

Figure 9.52:

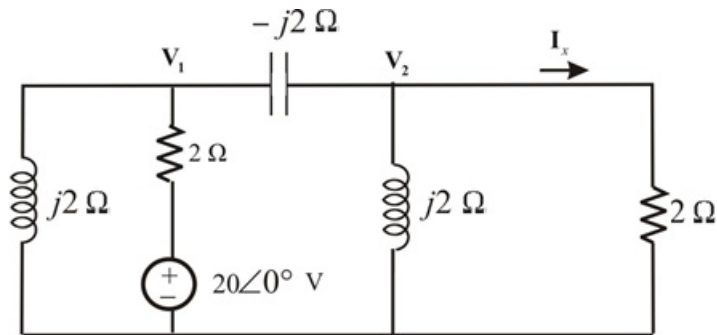


Step-by-step solution

Step 1 of 4

Refer to Figure 9.52 in the text book for the circuit.

Redraw the circuit by marking nodes, V_1 and V_2 .



Step 2 of 4

Apply Kirchhoff's voltage law at node V_1 .

$$\frac{V_1}{j2} + \frac{V_1 - 20\angle 0^\circ}{2} + \frac{V_1 - V_2}{-j2} = 0$$

$$V_1 \left(\frac{1}{j2} + \frac{1}{2} + \frac{1}{-j2} \right) + \frac{V_2}{j2} = \frac{20\angle 0^\circ}{2}$$

$$0.5V_1 - j0.5V_2 = 10\angle 0^\circ \quad \dots (1)$$

Apply Kirchhoff's Voltage law at node V_2

$$\frac{V_2}{j2} + \frac{V_2}{2} + \frac{V_2 - V_1}{-j2} = 0$$

$$V_2 \left(\frac{1}{j2} + \frac{1}{2} + \frac{1}{-j2} \right) - \frac{V_1}{-j2} = 0$$

$$-j0.5V_1 + 0.5V_2 = 0 \quad \dots (2)$$

Step 3 of 4

Write the equations (1) and (2) in matrix form.

$$\begin{bmatrix} 0.5 & -j0.5 \\ -j0.5 & 0.5 \end{bmatrix} \begin{bmatrix} \mathbf{V}_1 \\ \mathbf{V}_2 \end{bmatrix} = \begin{bmatrix} 10\angle 0^\circ \\ 0 \end{bmatrix}$$

$$\begin{aligned} \begin{bmatrix} \mathbf{V}_1 \\ \mathbf{V}_2 \end{bmatrix} &= \begin{bmatrix} 0.5 & -j0.5 \\ -j0.5 & 0.5 \end{bmatrix}^{-1} \begin{bmatrix} 10\angle 0^\circ \\ 0 \end{bmatrix} \\ &= \begin{bmatrix} 0.5 & -j0.5 \\ -j0.5 & 0.5 \end{bmatrix}^{-1} \begin{bmatrix} 10\angle 0^\circ \\ 0 \end{bmatrix} \\ &= \frac{1}{0.5} \begin{bmatrix} 0.5 & j0.5 \\ j0.5 & 0.5 \end{bmatrix} \begin{bmatrix} 10\angle 0^\circ \\ 0 \end{bmatrix} \\ &= \begin{bmatrix} 10 \\ j10 \end{bmatrix} \end{aligned}$$

The node voltage, $\mathbf{V}_2$ is $j10$ V.

Step 4 of 4

Determine the current through $2\ \Omega$ resistor.

$$\begin{aligned} \mathbf{I}_x &= \frac{\mathbf{V}_2}{2} \\ &= \frac{j10}{2} \\ &= j5\text{ A} \end{aligned}$$

Thus, the current through $2\ \Omega$ resistor is $j5\text{ A}$.